MODULE 25 · DOMAIN 5 - DATA COLLABORATION

Replication, failover, and the Domain 5 drill

Business continuity, then five scenarios closing the last domain.

DOMAIN 5 of 5 WEIGHT 10% MODULE 25 of 31

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Disaster recovery is **three** features, not one. Replicate, fail over, redirect the clients.

The edition boundary

FEATURE	EDITION
Database replication	Standard. Available to all editions
Replication of other account objects: warehouses, users, roles	Business Critical or higher
Failover and failback	Business Critical or higher
Client Redirect	Business Critical or higher

ON THE EXAM

This one line decides most questions here. You can replicate a database on Standard. You cannot fail over to it.

Replication groups and failover groups

- **A failover group is a replication group that can also fail over.** That is the entire difference.
- A **secondary** group in a target account is **read-only**. Promoting it to primary makes it read-write.
- You may promote **any target account** listed in the group's allowed accounts. Both provide **point-in-time consistency**.
- **Refresh applies only the changes since the previous refresh**, so cost scales with change rather than with size.
- Use **several groups** when some databases need refreshing more often than others.

> ON THE EXAM
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> Read-only until promoted is the fact that generates scenarios. A perfectly healthy secondary that rejects every write.

Practical constraints worth knowing

- After upgrading to Business Critical, **failover capabilities can take up to 12 hours** to become available.
- **Replicating to a lower-edition account raises an error by default**, so sensitive data cannot land somewhere weaker.
- Replication of a database is **blocked if it contains external tables**.
- Replication works across regions **within a Region Group**. Crossing Region Groups needs Snowflake Support.
- **Optimized refresh** scales refresh time with rate of change rather than object count. Newer, and awareness only.

> ON THE EXAM
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> The 12-hour delay is the one that catches people mid-incident. You cannot upgrade your way out of an outage this afternoon.

> A company on Standard Edition has been replicating a production database to a second account in another region for six months, and queries it successfully for reporting.
>
> A regional outage takes the primary account offline. The team attempts to promote the secondary database so writes can continue, and the statement fails.

Why, and what was needed?

A Failover requires Business Critical or higher. Database replication alone is Standard, and gives a read-only copy.

B The secondary must be refreshed immediately before promotion, and the primary is unreachable.

C Promotion requires the ORGADMIN role, which the team was not using.

D Secondary databases can never be promoted. A new primary must be built from a clone.

◆ PAUSE AND DECIDE

Why, and what was needed?

A Failover needs Business Critical. Replication alone is Standard.

The documented boundary. Standard replicates a database and gives a read-only secondary; promoting it is a Business Critical capability.

B A refresh is required before promotion.

Refreshing first is good practice for a planned failover. It is not what blocks this, and an unplanned failover accepts data loss up to the last refresh.

C Promotion requires ORGADMIN.

Account administrators manage failover groups. ORGADMIN handles organization-level tasks such as creating accounts.

D Secondaries can never be promoted.

Promotion is exactly what failover is. It simply requires the right edition.

WHERE THIS CAME FROM

Modules 2 and 25. Replicating for six months proves nothing about failover. They are separate features on opposite sides of the line.

> A team on Business Critical has a failover group replicating to a second region, refreshing every fifteen minutes. An analyst in the secondary region reports that they can query everything but every INSERT and UPDATE is rejected.
>
> There is no outage. The primary is healthy and serving production traffic normally.

What is happening?

A The refresh interval is too short, so the secondary is locked during each refresh.

B The analyst's role lacks INSERT and UPDATE on the replicated objects. Grants do not replicate.

C A secondary failover group is read-only. It becomes writable only when promoted to primary.

D Business Critical requires Tri-Secret Secure before secondary accounts accept writes.

◆ PAUSE AND DECIDE

What is happening?

A Locked during each refresh.

Refreshes do not lock the secondary against reads, and a 15-minute interval would not reject writes continuously.

B Missing grants. Grants do not replicate.

Grants on database objects are synchronised when roles are replicated. And the analyst can read everything, so their privileges are working.

C A secondary is read-only until promoted.

Documented. Read-only access is the defining property of a secondary; promotion is what makes it read-write.

D Tri-Secret Secure is required first.

Tri-Secret Secure concerns encryption keys. It has nothing to do with write access on a secondary.

WHERE THIS CAME FROM

Module 25. Nothing is broken. A secondary is a working read-only replica, and that is exactly what it is meant to be.

> A provider needs to give a consumer in another cloud region access to a database. They decide against a listing and set up database replication to a second account in the consumer's region, intending to share from there.
>
> The replication configuration fails. The database contains twelve tables, three secure views, and two external tables over an S3 data lake.

What is the most likely cause?

A. Secure views cannot be replicated, only shared. Convert them to tables first.

B. Replication is blocked when the database contains external tables.

C. Replication cannot be used to feed a share. Only listings support onward sharing.

D. A database can only be replicated if it contains fewer than ten objects per schema.

◆ PAUSE AND DECIDE

What is the most likely cause?

A. Secure views cannot be replicated.
They replicate normally. Secure views are also the standard way to share a filtered slice, so barring them would break sharing generally.

B. External tables block replication.
Documented: replicating a primary database is blocked if one or more external tables exist in it.

C. Replication cannot feed a share.
Replicate-then-share is the documented alternative to a listing for cross-region sharing.

D. A ten-object limit per schema.
Invented. There is no such limit.

WHERE THIS CAME FROM

Modules 24 and 25. The two external tables are the whole question. Twelve tables and three secure views are there to look like the problem.

A team creates a transient table to hold intermediate results, sets DATA_RETENTION_TIME_IN_DAYS to 0 to save cost, and drops it at the end of the pipeline. The next morning they discover the final step failed and they need the intermediate data back.

Someone suggests UNDROP, and failing that, contacting Snowflake Support for Fail-safe recovery.

What can be recovered?

A UNDROP will work, because it operates independently of the retention setting.

B The data is recoverable from the query result cache, since the pipeline ran within 24 hours.

C Fail-safe will recover it after 7 days, as it does for any dropped table.

D Nothing. Retention of 0 means no Time Travel, and transient tables have no Fail-safe.

◆ PAUSE AND DECIDE

What can be recovered?

A UNDROP works regardless of retention.

UNDROP relies on Time Travel. With retention at 0 there is no window, so there is nothing to restore.

B Recover from the result cache.

The result cache holds query results for reuse, not table contents for restoration. There is no way to restore a table from it.

C Fail-safe recovers it.

Fail-safe covers permanent objects only. Its absence is exactly why transient tables are cheaper.

D Nothing. No Time Travel, and transient has no Fail-safe.

Two independent reasons, and either alone is fatal. This is the cost of transient plus zero retention.

WHERE THIS CAME FROM

Module 23. Both choices were deliberate cost savings, and together they removed every recovery route. That is the trade, made explicit.

> A provider agrees to share data with a small partner who is not a Snowflake customer. They create a reader account, load the partner's users into it, and the partner begins running large analytical queries against the shared data.
>
> At the end of the month the provider receives an unexpectedly large bill and asks how to pass the compute cost to the partner.

What is the position?

A The provider creates and pays for reader accounts, including their compute. Costs cannot be passed through Snowflake billing.

B Reader account compute is billed to the consumer automatically once they accept the terms.

C The bill must be a Snowflake error, since reader accounts are read-only and read-only access is free.

D The provider should convert the reader account into a listing, which transfers compute cost to the consumer.

◆ PAUSE AND DECIDE

What is the position?

A The provider pays, and it cannot be passed through.

Reader accounts are created, owned and paid for by the provider. Resource monitors on the reader account are the practical control.

B Billed to the consumer.

That is the ordinary share model, where the consumer has their own account. A reader account belongs to the provider.

C Read-only access is free.

Read-only concerns writes, not cost. Every query consumes warehouse credits regardless of direction.

D Convert to a listing.

A listing needs the consumer to have a Snowflake account. This partner does not, which is why a reader account was chosen.

WHERE THIS CAME FROM

Module 24. The usual billing split does not apply here, because there is no consumer account. The provider owns both sides.

Where each question came from

QUESTION	MODULE TO REVISIT IF YOU MISSED IT
1. Promotion failed on Standard Edition	Modules 2 and 25 - the edition boundary
2. Secondary rejects every write	Module 25 - secondaries are read-only until promoted
3. Replication blocked by external tables	Modules 24 and 25 - replication constraints
4. Transient table with 0 retention, dropped	Module 23 - the retention grid
5. Reader account bill	Modules 11 and 24 - the provider pays

ON THE EXAM

Every question here spans two modules. Domain 5 is small enough that its facts interact constantly.

Domain 5 in five sentences

- **Time Travel is yours, Fail-safe is Snowflake's.** 90 days needs Enterprise, permanent, and somebody to set it.
- **Nothing is copied in a share.** Read-only, provider pays storage, consumer pays compute.
- **Direct share is one region. A listing crosses regions and clouds.** Reader accounts serve non-customers, and the provider pays.
- **Replication is Standard. Failover, failback and Client Redirect are Business Critical.** A secondary is read-only until promoted.
- **Clean room when neither party may see the other's rows.** Native app when your logic must run in their account.

> ON THE EXAM
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> Next: the confusion pairs module. Twenty things that look alike, gathered from all five domains.